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An IoT-Based Intelligent Framework for Real-Time Dam Monitoring and Control

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Abstract

Dams are critical infrastructures; their management and control are essential to ensure their safety, optimal utilization of water resources, and proactive maintenance. In this paper, we propose an innovative IoT-based intelligent framework for real-time dam monitoring and control. The system integrates a network of advanced sensors, edge devices, and cloud-based data analytics to enable real-time monitoring and predictive maintenance of dam infrastructures. The system utilizes ultrasonic water level sensors, temperature probes, and environmental sensors to collect critical data, which is then processed by microcontrollers equipped with integrated wireless connectivity. A key originality of our approach lies in the adoption of IoT design patterns, particularly the *Delta Update* and *Incremental Update* patterns, which significantly optimize data transmission, reduce bandwidth consumption, and improve energy efficiency. Data are securely transmitted via protocols to a cloud platform. Experimental results demonstrate the framework's capability to enhance the accuracy, responsiveness, and efficiency of dam monitoring. Our approach not only addresses the limitations of conventional monitoring techniques but also provides a scalable, pattern-driven solution that supports data-driven decision-making for improved dam safety and operational performance.

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1. Introduction

Dam management and control have become increasingly critical in the context of climate change, aging infrastructure, and growing demands for water resources. Traditional monitoring methods, mainly based on manual inspections and basic instrumentation, are often insufficient to provide the timely and accurate data required to prevent catastrophic failures and optimize water resource utilization. In contrast, the emergence of the Internet of Things (IoT) offers transformative potential by enabling continuous, real-time monitoring coupled with advanced analytics.

This paper presents an innovative IoT-based framework designed for real-time dam monitoring and control. Our solution integrates a network of advanced sensors, including ultrasonic water level sensors, temperature probes, and

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various environmental sensors, strategically deployed on dam infrastructure to capture critical parameters such as water level, temperature, and water quality. These sensors are interfaced with microcontrollers featuring integrated wireless connectivity, which aggregate and transmit the data via secure wireless protocols to a cloud platform.

A key originality of our approach lies in the adoption of IoT design patterns, specifically the *Delta Update* and *Incremental Update* mechanisms, to address the challenges of dam monitoring systems. IoT design patterns provide proven architectural solutions that facilitate the construction of reliable, reusable, and efficient systems [1]. Building upon established research in pattern-based IoT architectures [2, 3], we adapt and apply these specific patterns to optimize two critical aspects of dam monitoring: *communication efficiency* and *energy management*, which are crucial for resource-constrained environments.

In summary, our proposed framework addresses key challenges inherent in traditional dam monitoring systems, including delayed data acquisition, energy management issues, and limited remote accessibility. Using IoT technologies and pattern-driven data transmission strategies, the solution promises to revolutionize dam management and control, ultimately improving public safety and resource sustainability.

The remainder of this paper is organized as follows. Section 2 reviews the related work, Section 3 details the system architecture, Section 4 describes the implementation, experimental results, and discusses the efficiency of our proposed framework, and Section 5 concludes the paper and outlines future work.

2. State of the Art in Dam Management and Control

The management and monitoring of dam infrastructures have significantly advanced over recent decades. At first, dam control relied on manual inspections and basic instrumentation. Now, it has increasingly adopted digital and connected technologies. This section reviews existing approaches and highlights their strengths and limitations.

2.1. Traditional Methods

Conventional dam monitoring systems are often based on manual water level measurements, visual inspections, and mechanical sensors. These methods are simple to implement and cost-effective, making them widely adopted. However, they suffer from significant limitations such as low measurement frequency, human error, and a lack of real-time capabilities. Consequently, they are not suitable for early warning systems or dynamic dam operation.

2.2. Radar and Ultrasonic Systems

To overcome some of the negative aspects of manual inspections, radar and ultrasonic sensors have been adopted for contactless water level measurements [4]. Radar-based systems, whether pulse or FMCW (Frequency Modulated Continuous Wave), measure the distance to the water surface by emitting radio waves and timing their reflection; this method remains accurate even in fog, lowlight, or adverse weather conditions [5]. Such technologies enable more precise and continuous water-level readings and can be integrated into semi-automated monitoring systems.

However, radar sensors are generally expensive and require specialized calibration. Furthermore, they can be sensitive to environmental noise or surface reflections. This may limit their deployment in specific environments.

2.3. IoT-Based Monitoring Solutions

Smart monitoring systems incorporating a wide array of sensors connected via wireless protocols have been developed by recent advances in IoT. These systems enable continuous data acquisition and transmission to cloud platforms where data can be analyzed, visualized, and stored [6, 7]. Despite their advantages, these systems must address challenges such as energy autonomy and network reliability in remote areas.

2.4. Hybrid Architectures

A growing number of studies focus on hybrid architectures that combine IoT monitoring with digital twin models. These virtual representations of physical dam infrastructures allow simulation of dynamic behaviors in different

scenarios, supporting decision making and proactive maintenance strategies [8]. Although promising, digital twins require significant computational resources, high-quality data streams, and complex integration workflows, which may pose barriers to adoption in low-resource contexts.

2.5. Satellite-Based Monitoring Approaches

In addition to ground-based methods, satellite radar interferometry (InSAR) has emerged as a valuable tool for dam monitoring [9]. InSAR exploits radar signals from satellites to detect ground movements with millimeter precision by comparing phase differences between images acquired at different times. This technique enables large-scale, continuous monitoring that is essential for detecting potential structural instabilities and preventing catastrophic failures.

Unlike traditional geotechnical and geodetic monitoring, which require extensive instrumentation and manual data collection, InSAR provides a non-intrusive, efficient solution that covers vast areas with high temporal frequency. When integrated with traditional geotechnical and geodetic methods, InSAR significantly improves the reliability of dam health assessments. While geotechnical sensors provide localized soil and material data and geodetic techniques yield precise positional measurements, InSAR contributes wide-area deformation maps. This synergy enables more robust monitoring and enhances the capacity for predicting and mitigating risks.

2.6. Summary of Related Works and Comparative Analysis

The state of the art in dam monitoring and control reveals a clear evolution toward intelligent, real-time, and predictive systems. As illustrated in Table 1, while traditional approaches remain prevalent due to their simplicity and affordability, they are increasingly being complemented or replaced by IoT-enabled systems and digital modeling frameworks that offer superior responsiveness, scalability, and accuracy. Current IoT and hybrid approaches [6, 7] provide comprehensive monitoring capabilities, yet several critical challenges persist, particularly concerning energy efficiency, data transmission optimization, and systematic architectural reliability.

Our work distinguishes itself from existing solutions through its systematic integration of formally-verified IoT design patterns, while related studies focus on sensor deployment and data aggregation [6]. The use of IoT design patterns has been shown to facilitate the construction of reliable and reusable architectures. Our research line has consistently focused on the formal verification of such patterns. We initially formalized Service-Oriented Architecture (SOA) design patterns using the Event-B method to ensure correctness by construction [10]. This foundational work was then extended to the IoT domain, where we proposed a general formal approach to specify and verify IoT architectural patterns [1]. The practical effectiveness of this formal methodology was demonstrated in a concrete case study, where we modeled and proved IoT design patterns for an aged care solution [2]. Building directly upon this proven methodology, the present work applies and adapts the *Delta Update* and *Incremental Update* patterns [11, 12] to optimize communication efficiency and energy management in the specific context of dam monitoring systems, two major limitations in current IoT-based dam monitoring systems. These patterns optimize data transmission by reducing redundancy, improving bandwidth utilization, and minimizing energy consumption at the edge, which is crucial for resource-constrained environments.

3. System Architecture and Data Management

The proposed IoT-based solution for dam management and control relies on a multi-layered architecture that integrates IoT sensing, edge computing, robust communication, cloud-based analytics, and user decision support. The originality of our approach lies in the integration of **IoT design patterns**, particularly the *Delta Update* and *Incremental Update* mechanisms [11, 12], which optimize data transmission, reduce network load, and ensure energy-efficient operation. The overall architecture, shown in Fig. 1, enables real-time monitoring and proactive dam management.

3.1. Sensing Layer

The sensing layer consists of heterogeneous sensors deployed on and around the dam, including:

- **Water Level Sensors:** Ultrasonic and piezometric sensors for accurate depth measurements.

Table 1. Comparative overview of dam monitoring approaches

Approach	Advantages	Limitations
Traditional Methods	• Low initial cost and ease of deployment • Simple operation by on-site personnel	• Data collection is intermittent; lacks real-time monitoring • Susceptible to human error and subjectivity
Radar and Ultrasonic Methods	• High accuracy in water level measurement [4] • Potential for automated continuous monitoring	• High installation and maintenance costs • Requires specialized calibration and expertise [5]
Modern IoT Solutions	• Continuous, real-time data collection • Enables remote monitoring and predictive analytics via AI/ML [6, 7]	• Dependence on reliable connectivity in remote areas • Energy management and cybersecurity issues
Hybrid Approaches and Digital Twins	• Integrated, real-time view of dam performance • Supports proactive maintenance and optimization [8]	• Complex integration of heterogeneous data • High cost and need for multidisciplinary expertise
Satellite-Based Monitoring	• Provides millimeter-precision deformation maps • Large-scale, continuous, and non-intrusive monitoring • Standardized datasets available (e.g., EGMS) • Effective when combined with geotechnical/geodetic methods	• Requires satellite data processing expertise • Limited by revisit frequency and atmospheric effects • Less effective for capturing rapid, short-term variations [9]
Our proposed IoT Framework	• Real-time monitoring, high accuracy and low latency • Energy and bandwidth efficient patterns • Scalable and pattern-driven architecture • Human decision support with alert systems • Cost-effective compared to radar/satellite solutions	• Connectivity dependency in remote areas • Requires periodic sensor calibration • Energy autonomy challenges for long-term deployment • Security risks in public network transmission



Fig. 1. Multi-layer architecture for Dam management and control integrating IoT design patterns

- **Temperature Sensors:** Sensors used to monitor water temperature and detect risks of evaporation.
- **Environmental Sensors:** Devices to capture rainfall and humidity, providing complementary hydrological and structural information.

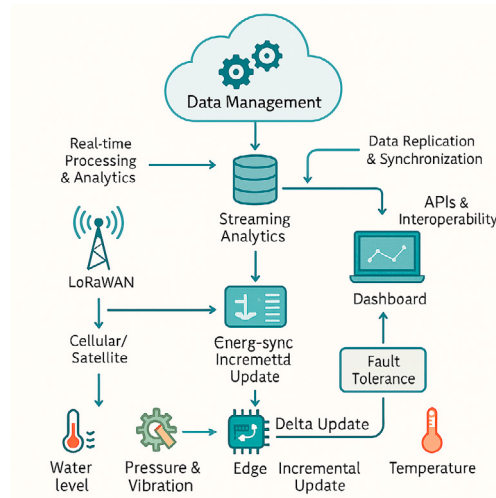


Fig. 2. Data flow and communication management

3.2. Edge and Communication Layer

At the edge, microcontrollers (NodeMCU) aggregate data, perform lightweight analytics, and transmit results to the cloud. A key innovation of this layer is the adoption of **IoT design patterns** for data transmission [11, 12]. Two main patterns are used. The first one is the **Delta Update** pattern. By applying this pattern, only the variations between the current and previous sensor readings are transmitted, reducing bandwidth and energy consumption. The second pattern is the **Incremental Update** pattern. By applying this pattern, only newly generated data since the last transmission is forwarded, avoiding redundancy, enabling efficient synchronization, and improving resilience against temporary disconnections. These mechanisms ensure efficient, adaptive communication, even in remote areas with limited connectivity.

3.3. Cloud and Data Analytics Layer

In the cloud, sensor data streams are securely ingested and processed in real time:

- **Data Management:** Time-series databases store high-frequency data, while relational databases manage meta-data.
- **Analytics:** Machine learning models (e.g., anomaly detection, time-series forecasting) exploit the reduced and filtered data enabled by Delta and Incremental Updates to forecast risks such as overflow or structural anomalies.
- **Security and Integrity:** End-to-end encryption, authentication, access control, and checksum validation safeguard data, while automated backups ensure reliability and compliance.

The mechanisms of data acquisition, preprocessing, transmission, and cloud integration are summarized in Fig. 2. This diagram highlights the end-to-end workflow of data flow and communication management, from IoT sensors at the edge to secure storage and analytics in the cloud.

3.4. User Interface and Decision Support Layer

The processed information is delivered to dam operators via a web-based dashboard. It provides real-time visualization of water levels, temperature, and structural stress indicators. The system automatically generates alerts based on thresholds, with notifications sent via SMS and email.

3.5. Integrated Workflow

The workflow connects sensing, preprocessing at the edge, optimized communication, and cloud analytics into a unified decision-support system. By embedding Delta and Incremental Update patterns into the architecture, the solution reduces redundancy, minimizes bandwidth usage, and improves scalability and energy efficiency. This pattern-driven design improves the robustness of dam monitoring and supports proactive data-driven management of critical water infrastructure.

4. Implementation and Experimental Results

This section presents the practical implementation of the proposed IoT-based dam management solution and the evaluation of its performance in both simulated and pilot field conditions. The implementation covers the hardware and software setup, system integration, and experimental validation of the Delta and Incremental Update mechanisms.

4.1. Hardware and Software Setup

The prototype system was built around a NodeMCU microcontroller due to its built-in Wi-Fi capability and compatibility with various sensors. The deployed sensors included:

- **Water Level Sensor (Solu SL067):** Installed in the reservoir to measure the water level as a percentage of full capacity.
- **Temperature Sensor (DS18B20):** A sensor to monitor water temperature and detect changes that affect the risks of evaporation.
- **Ultrasonic Distance Sensor (US-100):** Used to confirm water level readings by measuring the distance from the water surface.

The sensors were connected to the NodeMCU and programmed using the Arduino IDE. We used libraries to ensure reliable communication. Sensor data were transmitted securely to the Ubidots cloud platform using MQTT, where they were stored in a database and visualized in real time through a dedicated dashboard.

In Fig. 3, we show the system's ability to visualize data in real time. It shows the dashboard widgets implemented on the Ubidots platform. This interface enables operators to continuously monitor water level, temperature, and other parameters, while also receiving alerts in case of anomalies. This visualization is essential to improve situational awareness and support proactive dam management.

In addition to visualization, the efficiency of data transmission is a crucial element of the system's performance. Fig. 4 illustrates the impact of the Delta Update mechanism on temperature data transmission, showing a significant reduction in redundant values while maintaining accuracy. This optimization ensures the more efficient use of bandwidth and energy resources, which is vital for deployments in remote environments.



Fig. 3. Dashboard widgets for real-time data visualization in Ubidots

4.2. System Integration and Testing

After assembling the setup, the integration of the system involved several steps. The first step is *Sensor Calibration*, in which each sensor was tested under controlled conditions to ensure accuracy. The second step is *Edge Processing*. Preprocessing on the NodeMCU applied filtering and anomaly detection. Using the *Delta Update* mechanism, only significant variations were transmitted, reducing communication overhead. Finally, the third step is *Cloud Processing*. It was implemented with real-time dashboards, enabling anomaly detection and automatic alerts whenever thresholds were exceeded. The system was first tested in a laboratory environment that replicated dam conditions, and then validated in a small-scale pilot deployment.

4.3. Experimental Results and Discussion

The experiments assessed measurement accuracy, transmission latency, and data reduction performance. Results are summarized in Table 2.

Table 2. Summary of Experimental Results

Parameter	Result
Water Level Measurement Accuracy	93–95%
Temperature Measurement Accuracy	93%
Transmission Latency	1.5–2 s
Data Reduction (Delta Update)	75%
Data Reduction (Incremental Update)	60%
System Availability	99%

The effectiveness of the Delta Update mechanism is illustrated in Fig. 4, which compares the transmission of temperature data with and without the application of this mechanism. The figure illustrates a significant reduction in redundant data, resulting in optimized bandwidth utilization while maintaining measurement accuracy.

With these results, we can confirm that our proposed system supports real-time monitoring with low latency, accurate measurements, and significant bandwidth and energy savings due to design patterns used for optimized data transmission. The evaluation demonstrates that the IoT-based framework effectively combines sensor data, edge processing, and cloud analytics for proactive dam monitoring. Key strengths of our system include:

- **Reliability:** Continuous monitoring with high system availability (99%).
- **Efficiency:** Significant reduction in data volume while preserving accuracy.
- **Proactivity:** Alerts triggered within 2 seconds during simulated flood events.

Nevertheless, challenges remain in ensuring robust connectivity in remote environments, maintaining sensor calibration, and optimizing energy consumption for long-term deployments. Future work will focus on proposing predictive models, strengthening network resilience, and scaling the system to larger infrastructures.

Experiments have shown that our proposed IoT-based framework for dam management and control is a significant improvement over traditional monitoring methods. The system's ability to provide real-time data with low latency and high accuracy allows for proactive decision-making, which is vital for safeguarding the safety and efficiency of dam infrastructures. This proactive approach is crucial, especially in scenarios involving rapidly changing environmental conditions or imminent flood risks.

However, several challenges remain. The system's performance is highly dependent on reliable wireless connectivity. This can be an obstacle in remote or harsh environments. Additionally, while the sensor accuracy is within acceptable limits, maintaining sensor calibration and ensuring sustained energy supply for battery or solar-powered nodes requires further consideration. Another critical aspect is security because the transmission of sensitive data over public networks requires robust encryption protocols. Overall, our proposed solution shows great potential in revolutionizing dam monitoring practices. By enabling remote and real-time, it lays the foundation for more resilient and responsive water resource management strategies.

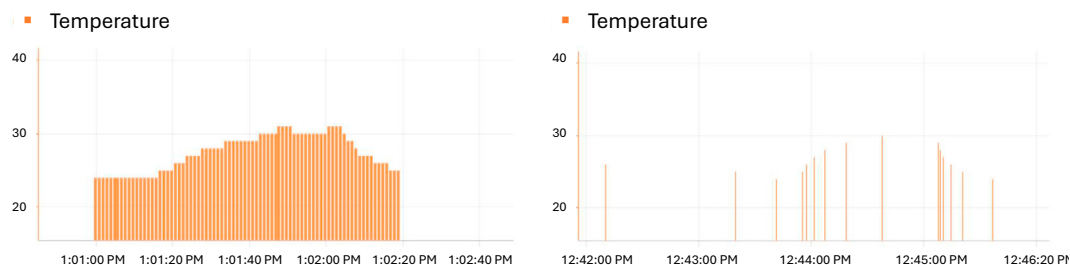


Fig. 4. Temperature transmission with and without Delta Update

5. Conclusion and Future Work

In conclusion, this paper has presented an IoT-based intelligent framework for dam management and control. The system integrates real-time sensor monitoring and edge processing to enhance the safety, reliability, and efficiency of dam operations. Our system's ability to deliver timely data with high accuracy is demonstrated by the experimental results, which enable early anomaly detection and proactive decision-making. These improvements are a significant improvement over traditional dam monitoring techniques. Despite these promising outcomes, several challenges remain. The performance of the system depends on reliable wireless connectivity, which can be difficult in remote environments. Energy management for sensor nodes and ensuring robust data security also require further attention.

Future Work will focus on adding predictive models by incorporating a range of hydrological and environmental variables, as well as exploring advanced machine learning algorithms.

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